Showcasing Robots 📣

Cadette Robotics Badge — Girls Build Bots

Presented by your robotics mentors!

The robotics ecosystem 

FIRST programs by age: - **FIRST LEGO League** (9–14) — LEGO robots, research projects - **FIRST Tech Challenge** (12–18) — metal robots, Java programming - **FIRST Robotics Competition** (14–18) — 120-lb robots, 6-week build season

Plus: VEX Robotics, science fairs, hackathons, maker spaces…

Live demo: our FRC robot 🤖

Teardown talk — watch for: - Drive train and motors - Sensors (encoders, cameras, gyro) - The manipulator (our robot’s “tool”) - Where the code lives

Q&A as we go — interrupt us!

Teams need more than builders 🧩

Real FRC team roles: - Programming • CAD/design • Machining - **Strategy** — analyzing matches, picking alliances - Scouting & data analysis 📊 - Business, sponsorships, media - **Outreach** — which is literally why we’re here today

Robotics careers work the same way: mechatronics, CS, biomedical engineering, product design, technical writing…

Your mission: the public pitch 🎤

3 minutes. Audience: **non-technical** — imagine pitching a city council for funding.

1. **Hook** — the human story: who needs this and why
2. **Solution** — your robot, ZERO jargon
3. **Evidence** — what your testing and iteration showed
4. **Ask** — what you'd need to build the real thing

Kill the jargon ⚔️

Don’t say	Say
“actuator”	“the motor that moves the arm”
“failure-mode analysis”	“we figured out what breaks first”
“iterate”	“we tested it and made it better”

Mentors will flag jargon during rehearsal. **If your grandmother wouldn’t get it, rewrite it.**

Rehearsal rules 🎬

- **Two full run-throughs**, timed
- Every team member has a speaking role
- Feedback round after each: clarity, pacing, eye contact
- Nervous is normal — preparation beats nerves every time

Showtime 🌟

Present to the audience. Then: **Q&A**.

Handling questions like a pro: - Pause. Breathe. It’s okay to think. - Don’t know? *“Great question — we’d need to test that.”* - Disagreement? *“That’s a fair point — here’s our reasoning…”*

What you accomplished ✅

- Learned the full robotics ecosystem and career paths
- Translated technical work for a general audience
- Delivered a persuasive public pitch
- Handled live Q&A
- **Completed the entire Cadette Robotics badge series** 🥳

Your next step — pick ONE 🚀

- Join or start a **FIRST Tech Challenge** team → firstinspires.org
- **Girls Who Code** clubs & free summer programs → girlswhocode.com
- **Technovation Girls** — build an app, pitch it globally → technovation.org
- NASA internships start in high school → intern.nasa.gov
- **Talk to us after** — here's our team's contact info. Seriously. We started exactly where you are.